GIK INSTITUTE
OF ENGINEERING SCIENCES AND TECHNOLOGY

Elevator Control System

A Report for DLD Semester Project

Prepared by:

Name	Registration ID	Email
Wajeha Umer	2024659	u2024659@giki.edu.pk
Syeda Fatima tu Zahra	2024625	u2024625@giki.edu.pk
Syed Rayyan Hasan	2024621	u2024621@giki.edu.pk
Aisha Noor	2024075	u2024075@giki.edu.pk

Prepared for:

CE221L: Digital Logic Design Lab

Waheed Iqbal

waheed.iqbal@giki.edu.pk

Submission Date: 21st November 2025

Elevator Control System

Wajeha Umer
BS CS of FCSE
2024659
u2024659@giki.edu.pk

Syed Rayyan Hasan
BS CS of FCSE
2024621
u2024621@giki.edu.pk

Aisha Noor
BS CS of FCSE
2024075
u2024075@giki.edu.pk

Syeda Fatima Tu Zahra
BS CS of FCSE
2024625
u2024625@giki.edu.pk

***Abstract*— This project implements a four floor elevator controller using sequential and combinational logic and a queue scheduling mechanism. Floor requests are encoded into a two bit value and stored in a four slot queue implemented using three parallel shift registers. The elevator always serves the request at the front of the queue. A comparator determines whether the elevator should move up, move down, or stop when the target floor is reached. The elevator halts on arrival and waits for a manual-dequeue action, after which the queue shifts and the next request becomes active. Proteus simulation is used to verify correct queue ordering and elevator motion.**

Keywords—elevator controller, queue scheduling, FIFO, shift register, multiplexer, digital logic design, Proteus

I. INTRODUCTION

Elevator control systems are a common example used to demonstrate the application of digital logic design in real-world problems. Such systems require correct handling of multiple user requests, decision-making based on current system state, and reliable control of motion. In digital systems, these requirements can be implemented using combinational and sequential logic elements such as gates, flip-flops, counters, comparators, and shift registers.

One important aspect of an elevator controller is the method used to schedule multiple floor requests. In this project, a queue-based (FIFO) scheduling mechanism is used, ensuring that floor requests are served strictly in the order in which they are received. This approach avoids request starvation and provides deterministic and predictable system behavior.

The purpose of this project is to design and simulate a 4-floor elevator controller using discrete 74xx TTL integrated circuits, without the use of microcontrollers or programmable logic. The system stores incoming requests in a hardware-implemented queue, selects the front request as the current target, and controls elevator movement accordingly. Proteus simulation software is used to design, test, and verify the correct operation of the logic before any hardware implementation.

II. PROJECT DESCRIPTION

The proposed system is a 4-floor elevator controller utilizing First-In-First-Out (FIFO) methodology to manage floor requests sequentially. The controller accepts floor requests, stores them in a queue, and controls elevator movement so that requests are served strictly in the order they are received. All functionality is implemented using combinational and sequential logic circuits, and no programmable device.

A. Request Handling and Encoding

Floor buttons 0–3 trigger a combinational logic block that encodes the physical press into a 2-bit binary value representing floor number. A request-valid pulse is simultaneously generated to push the data into the queue. The encoding logic ensures that each request is converted into a valid binary form before being processed by the queue.

B. Queue Implementation

The request queue is implemented as a 4-slot hardware queue using three parallel shift registers. One register stores the validity of each slot, while the other two registers store the most significant bit and least significant bit of the floor number. Each slot in the queue therefore represents one complete request. When a new request is received, the control logic identifies the first empty slot in the queue and stores the encoded floor number in that position. The validity bit of that slot is set, while all previously stored requests remain unchanged. This mechanism guarantees first-in-first-out behavior and prevents overwriting of existing requests.

C. Target Selection

At any time, the elevator considers only the request stored at the front of the queue. The floor number stored in the first slot is treated as the current target floor. If the queue is empty, no movement occurs.

D. Arrival and Stop Mechanism

When the elevator reaches the target floor, a stop signal is generated which also serves as door-open operation. This signal halts further elevator movement and holds the elevator at the current floor. Unlike fully automatic systems, this design does not immediately remove the request from the queue upon arrival.

E. Manual Dequeue Operation

After arriving at the target floor, a manual dequeue action is required which also serves as the door closing operation. This action shifts the queue forward by one position, removing the served request and making the next request active. Once the dequeue operation is completed, the stop condition is cleared, and the elevator resumes operation if additional requests remain in the queue.

III. PROJECT REQUIREMENTS

The following requirements were defined for the design and implementation of the system:

A. Operational logic and Scheduling

- The system shall accept multiple floor requests and store them in a queue; in the order they are received.
- The elevator shall always serve requests at the front of the queue.
- A manual dequeue (door close) operation shall be required to remove the served request from the queue.

B. Hardware and Control Constraints

- The queue shall be implemented using TTL logic components, without the use of microcontrollers or programmable devices.
- The elevator shall be able to move upward, downward, or stop based on the comparison between the current floor and the target floor.
- The elevator shall stop automatically when it reaches the target floor.

C. Simulation and Verification

- The system should be simulated and verified using Proteus design suite software.

IV. BLOCK DIAGRAM

The overall system can be divided into several functional blocks, each responsible for a specific task in the elevator controller. The overall system architecture is illustrated in the block diagram provided in Fig. A1

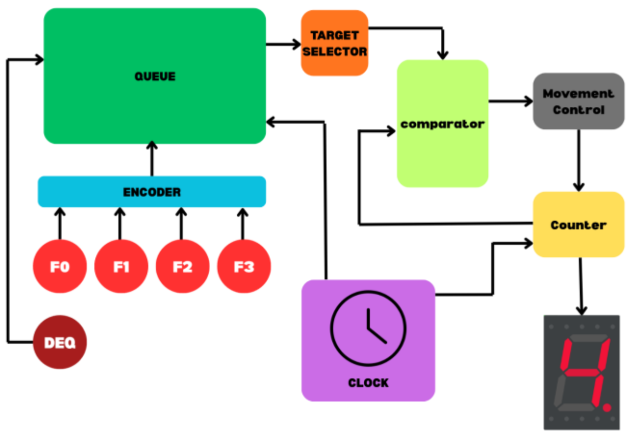

Fig. A1. Block diagram of the 4-floor elevator controller showing request encoding, FIFO queue scheduling, target selection, comparison logic, and elevator movement control.

V. HARDWARE COMPONENTS SPECIFICATION

The elevator controller is implemented using standard 74xx series TTL integrated circuits along with basic input and display components. The major hardware components used in the design are listed below.

- 74LS194: 4-bit bidirectional shift register.
- 74LS157: Quad 2-to-1 multiplexer
- 74LS85 – 4-bit magnitude comparator
- 74LS193 – Up/Down binary counter
- 74LS74 – Dual D-type flip-flop
- 74LS08 – Quad 2-input AND gate
- 74LS32 – Quad 2-input OR gate
- 74LS04 – Hex inverter
- 74LS47 – BCD to seven-segment decoder
- Push Button Switches
- LEDs – Logic and queue visualization
- Seven Segment Common Anode Display
- +5V DC Power Supply

VI. PROTEUS SCHEMATICS

The complete elevator controller was designed and simulated using Proteus 8. The circuit schematic includes all logic components required for request handling, queue implementation, elevator movement control, and display.

The queue is implemented using shift registers and multiplexers, while the elevator movement logic is implemented using counters, comparators, and control gates. Input push buttons are used to generate floor requests and manual dequeue operations. A seven-segment display is used to indicate the current elevator floor, and LEDs are used to visualize the queue contents.

The Proteus schematic was used to verify correct working before any hardware implementation. The complete circuit schematic is included in the Appendix.

VII. REFERENCES

[1] T. L. Floyd, Digital Fundamentals, 11th ed. Pearson Education, 2015.

[2] Texas Instruments, “74LS194 Bidirectional Universal Shift Register Datasheet.”

[3] Texas Instruments, “74LS157 Quad 2-to-1 Multiplexer Datasheet.”

[4] Texas Instruments, “74LS85 4-Bit Magnitude Comparator Datasheet.”

[5] Texas Instruments, “74LS193 Up/Down Binary Counter Datasheet.”

IX. APPENDIX: PROTEUS SCHEMATICS

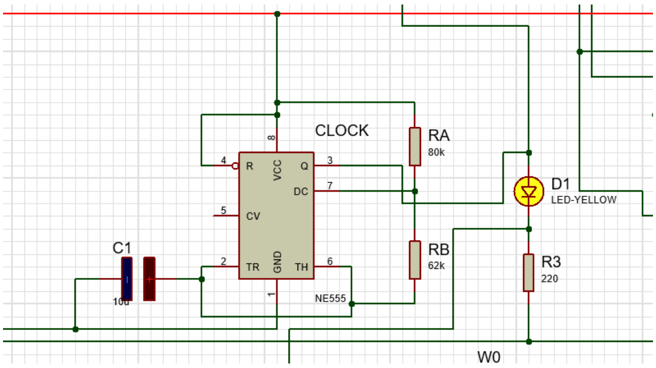

Fig. B1. The System Clock

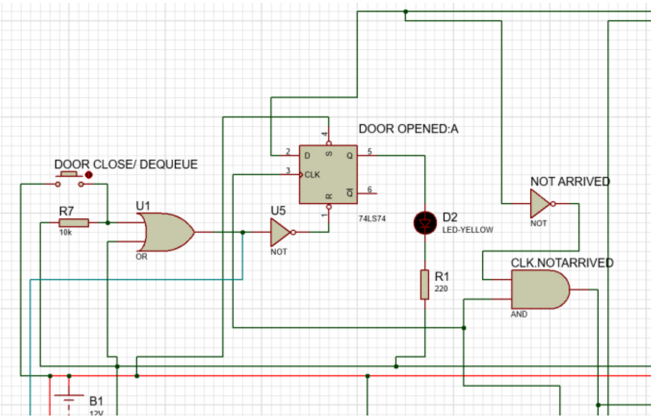

Fig. B2. The Dequeue and Door Close Operation

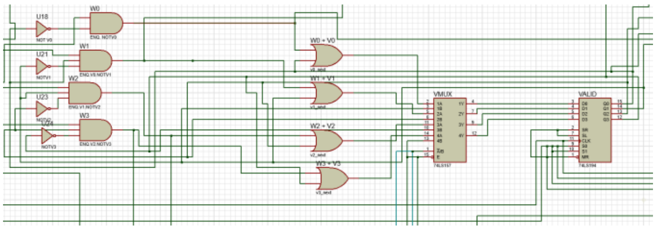

Fig. B3. Encoder of the validation bit and the register to store it

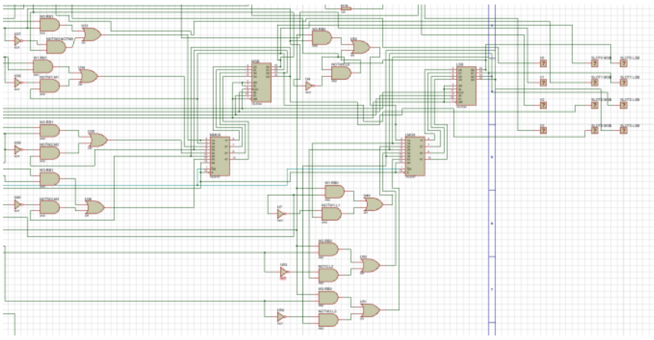

Fig. B4. Encoder of the MSB and the LSB and the registers to store them.

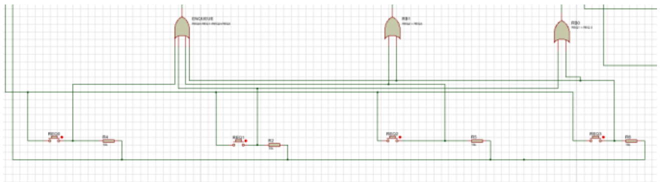

Fig. B5. The Request Logic

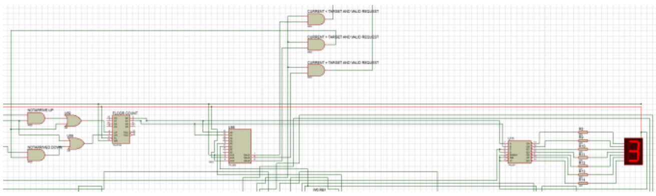

Fig. B6. The Movement Control Logic

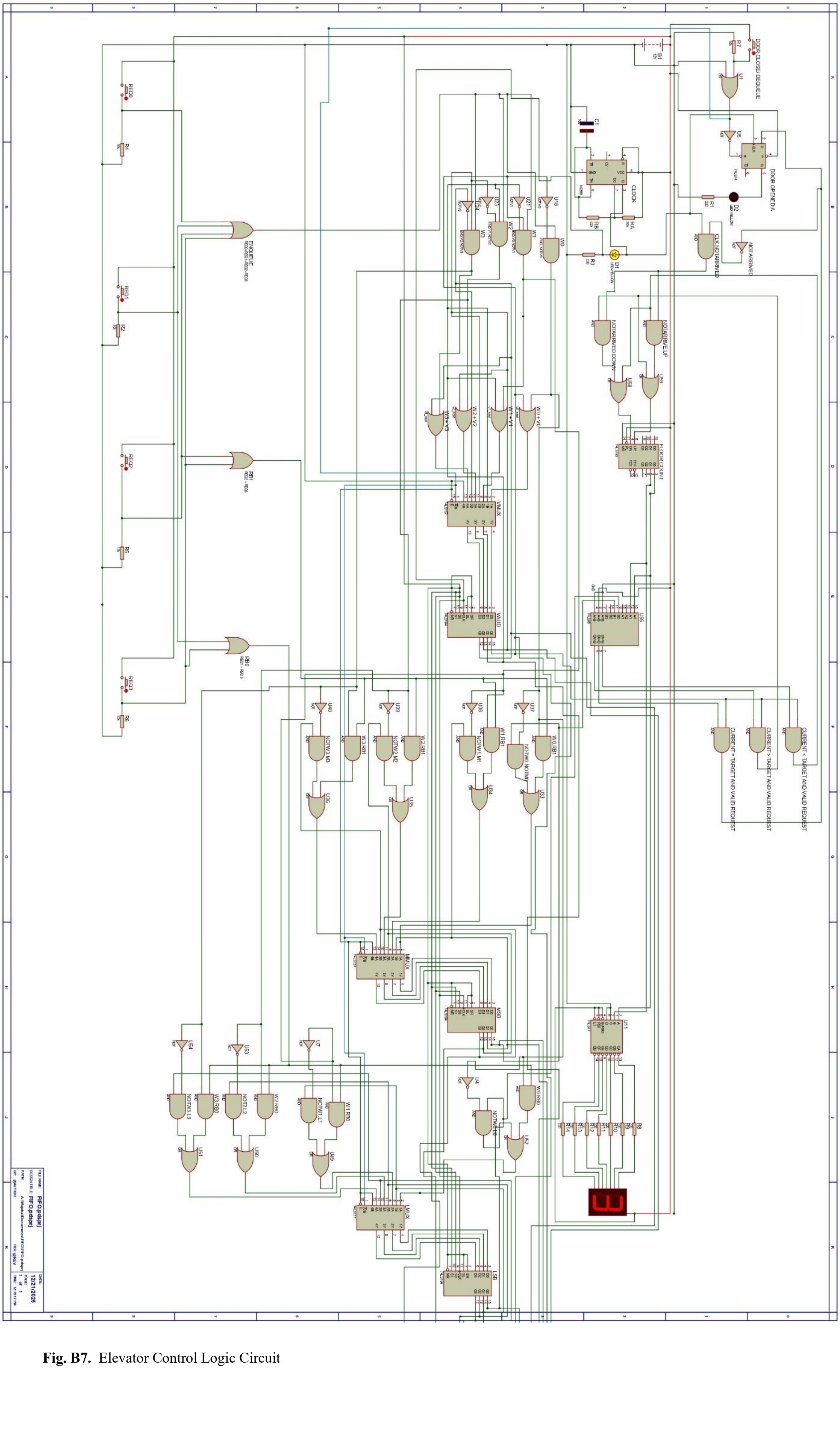

Fig. B7. Elevator Control Logic Circuit